University of Cape Town
Department of Computer Science

Computer Science CSC3003S, 2024

November Exam

Paper 2 – Theory of Algorithms and Computation

Answer ALL questions in both sections.

Each section MUST be written in **separate books**. Given your two books, please **write TOA at the top of the book** you will use for Section A: TOA and **write TOC at the top of the book** you will use for Section B: TOC. If you use additional books, please label them also and place these inside the first book for the relevant section.

Marks: 50 *Marks per question are shown in brackets.*

Time: 90 minutes + 10 minutes reading time

The use of calculators is permitted.

Read through the entire exam paper before you start writing so that you can answer the easy questions first. The relative importance of the questions is indicated by the marks that are allocated to them. Reading time of **10 minutes** is included in the allocated time above.

Section A. TOA – Theory of Algorithms [25]

Please write this section in its own book. Write TOA at the top of the cover page of the book. This section includes Questions 1-3.

Question 1. **[5]**

a) State whether the following statement is TRUE/FALSE: We say that f(n) ∈ O(g(n)) if there exists some constant k and positive integer n_i such that f(n) <= kg(n) for all n_i <= n . [1]

b) Consider the following algorithm and answer the questions that follow:

```
ALGORITHM A1(A[0...n-1]):
  for i ←1 to n-1:
    if A[i] mod 2 == 0:
        A[i-1] = 0
  return A
```

i. Explain what algorithm *A1* does using 1-3 sentences. [1]

ii. What is the basic operation? [1]

iii. What is the asymptotic computational complexity of the algorithm in BigTheta notation? (Show all steps that lead to answer). [2]

Question 2. **[10]**

You are tasked with creating an algorithm to sort an array of arbitrary size where each item can take on only one of three possible values 180, 142, and 140.

a) If you were to sort the array [180, 142, 142, 180, 140] using Quicksort with a constant divide factor of 2, relying on Lomuto partitioning, how many items will be in the sub-arrays after splitting the input array for the first time? Support your answer. [4]

b) In the best case, what is the efficiency of the Quicksort algorithm described above? Support your answer. [2]

c) Outline an algorithm (in English or pseudo-code) to solve this problem in a manner that is more efficient than Quicksort by trading-off space for time. [4]

Question 3. **[10]**

You are given a collection of N coins, each has a specified diameter and denomination, and you are tasked with selecting a subset that is most of valuable and if you were to line them up in a row (in a perfect straight line with no space in between) then they cannot exceed a length of 40 mm. There is only once instance of each coin.

Coins: (20mm, R10), (30mm, R20), (22mm, R15), (4mm, R5)

a) Create an exhaustive search algorithm (provide pseudo-code) for solving the task. [4]

b) Create a dynamic programming algorithm (provide pseudo-code) to solve the task.

i. Specify the recurrence relations for the dynamic programming solution. [2]

ii. Specify an algorithm that uses the recurrence relations to solve the problem. [2]

c) Contrast the two algorithms from Q3a and Q3b and explain whether, and why, if applicable, one of them is more efficient. Support your answer. [2]

PAPER CONTINUES ON NEXT PAGE

Section B. TOC – Theory of Computation [25]

Please write this section in its own book. Write TOC at the top of the cover page of the book. This section includes Questions 4-10.

Question 4. **[2]**

Recall the formal definition of a Turing machine and variations of them that we have seen. Then answer the following questions with either a 'yes' or a 'no'.

a) Is the tape alphabet Γ necessarily the same as the input alphabet Σ?

b) May a Turing Machine, upon processing the input, halt in a state that is not in the set of final states F?

c) Is a Turing Machine with multiple tapes more expressive than a Deterministic Turing Machine with one tape?

Question 5. **[4]**

Describe in *at most one sentence* the language that the following TM accepts. Write it in the form of, e.g., the language of strings such that "each input *w* contains twice as many 0s as 1s" (*not* a discursive story of an individual run such as 'when it sees a 1, then…').

Note: B is the blank symbol.

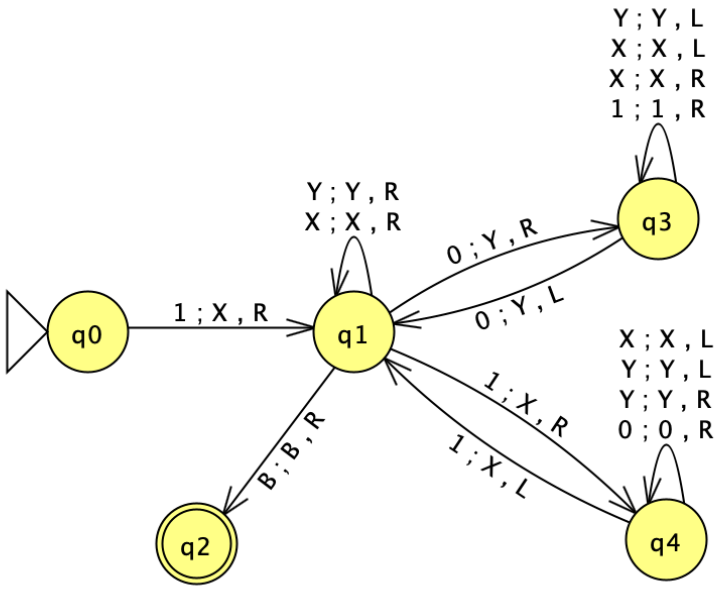


Figure 1 for Question 5

Question 6. **[4]**

Consider the following TM, which should add 1 to a number stored in a byte (in binary). For instance, it should be able to convert 01111111 into 11111111 on the tape, and 10101010 into 10101011, or reject if we have 256 (i.e., 11111111). You may assume the byte is preceded by a "." (as if it were part of an IP address, say). The state diagram below in Figure 2 is faulty, however. Which state(s) or transition(s) need to be added/removed/modified such that it will conform to the requirements, assuming valid input? Select the best option from the five options listed below the figure.

Note: B is the blank symbol.

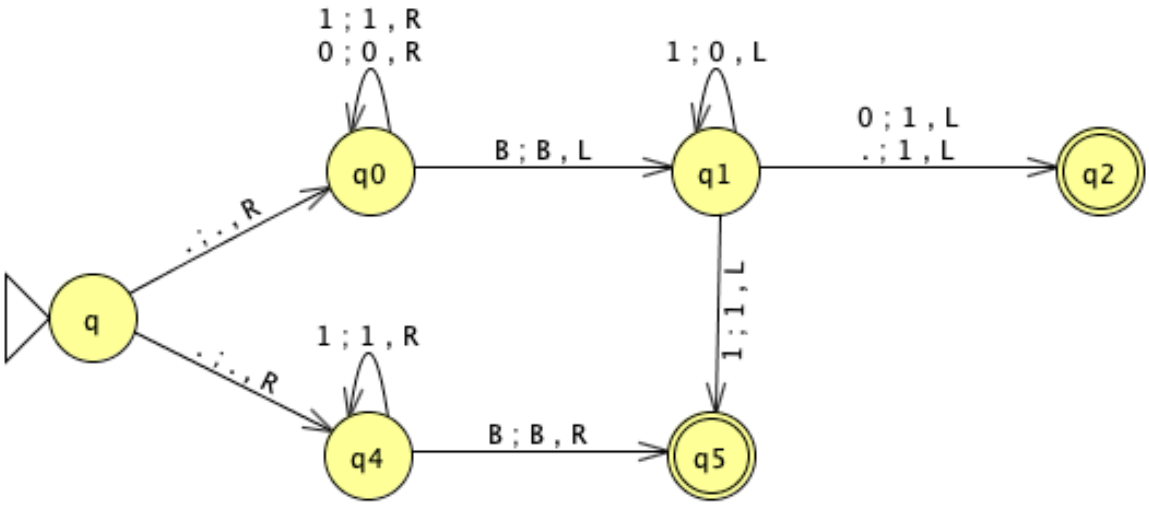


Figure 2. Figure for question 6.

a) Remove q5 and its two incoming transitions.
b) Modify δ(q1,.) = (q2, 1, L) into δ(q1,.) = (q2, ., L), and make q0 the start state instead of state q.
c) Remove the whole lower branch, i.e., remove δ(q, .) = (q4, ., R), δ(q4, 1) = (q4, 1, R), δ(q4, B) = (q4, B, R), q4, and q5, and thus also δ(q1, 1) = (q5, 1, L).
d) Remove δ(q1,1) = (q5, 1, L), either modify δ(q1,.) = (q2, 1, L) into δ(q1,.) = (q2, ., L) or remove it altogether, and make q5 a non-final state.
e) Add δ(q4,0) = (q4, 0, R), reverse δ(q1,1) = (q5, 1, L) into δ(q5,1) = (q1, 1, L), and modify δ(q1,.) = (q2, 1, L) into δ(q1,.) = (q2, B, L)

Question 7. **[3]**

The Universal Turing Machine (UTM) with its language L_U and the diagonalization language L_d were crucial to proving the limits of computation. Which of the following statements are true about that?

a) The principle of diagonalization only applies to L_d where the Turing Machine is encoded on 0s and 1s, but cannot be reused for some other languages to show they are also not recursively enumerable.
b) Since L_d is not recursively enumerable, we can use it to show some property of L_U, such as it being not recursive.
c) L_U takes as input a binary encoding of a Turing Machine together with some input string *w*.
d) L_d is the language such that it disagrees with each $L(M_i)$ on at least one string w_i, i.e., no Turing Machine can accept L_d.
e) L_U consists of those sets of strings that encodes the TM for any program code that has a function that, once it terminates computation, prints "Hello, World" as its first 12 characters.
f) The UTM is defined by halting, rather than by reaching a final state, and thus is an appropriate TM to illustrate the halting problem.

Question 8. **[3]**

There are variations of the simple Deterministic Turing Machine. Of the following modifications to Turing Machine M with some problem P to solve, which one(s) make(s) P a decidable problem, whatever M needs to do, given some suitable input *w*? You may pick more than one answer.

a) Restrict it such that each tape cell can change its symbol at most once.
b) Testing whether M terminates on *w* using at most *k* cells of the tape, where *k* is an integer number.
c) Limiting the tape symbols to only 0, 1, X, Y, and blank (B).
d) Limit the number of moves that M can make when trying to solve P to n^{15} moves, on input *w* that is of length *n*.

Question 9. **[3]**

Which one of the following problems is undecidable? Select the best option and state in at most one sentence why.

a) Computing the knapsack problem, i.e., "is there a subset of the items whose cumulative weight matches *k*?" where *k* is the capacity of the 'knapsack'.
b) Is the Boolean formula α satisfiable? (α may be, e.g., $(p \wedge q) \vee \neg r)$)
c) Does the language for Turing Machine M contain or print any palindromes?
d) Does there exist some input *w* for Turing Machine M such that it makes at least 7 moves?
e) Is the length of the string of the language of Turing Machine M greater than or equal to 0?

Question 10. **[6]**

There are seven languages (or equivalently, problems) A, B, C, D, E, F, and G. All we know about them is the following:

- A is in P.
- B is NP-complete.
- C is in NP.
- D is Recursive.
- E is Recursively Enumerable.
- F is Recursively Enumerable but not Recursive.
- G is non-Recursively Enumerable.

Of the four statements below, select any TWO (2), state for each of the selected two whether they are true or false, and justify in one sentence each why.

a) There may be a reduction from D to G.
b) There certainly is a polytime reduction from F to A.
c) There may be a polytime reduction from B to C.
d) There certainly is a reduction from D to B.

Appendix A – Master Theorem

If we have a recurrence of this form: $T(n) = aT(n/b) + f(n)$, $f(n) \in \Theta(n^d)$ and $T(1) = c$ then

$T(n) \in \Theta(n^d)$ if $a < b^d$

$T(n) \in \Theta(n^d \log n)$ if $a = b^d$

$T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$

- END -